

Recommendations — cca-mets-scope-unknown-z3p9

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Profile snapshot

- **Primary site:** biliary tract (cholangiocarcinoma)
- **Histology:** cholangiocarcinoma (adenocarcinoma of the biliary epithelium); intrahepatic vs perihilar vs distal/extrahepatic subsite not specified at intake
- **Stage:** IV (metastatic) per the stated diagnosis; full AJCC TNM staging and metastatic-site inventory not supplied at intake
- **ECOG:** 1
- **Age band:** 60-69
- **Sex:** unknown
- **Biomarkers:** FGFR2 fusion/rearrangement unknown (not yet tested) (ngs_pending); IDH1 R132 mutation unknown (not yet tested) (ngs_pending); HER2 (ERBB2) amplification/overexpression unknown (not yet tested) (ngs_pending); BRAF V600E unknown (not yet tested) (ngs_pending); MSI-H / dMMR unknown (not yet tested) (ngs_pending); Tumor mutational burden (TMB) unknown (not yet tested) (ngs_pending); NTRK1/2/3 fusion unknown (not yet tested) (ngs_pending); RET fusion unknown (not yet tested) (ngs_pending); KRAS G12C unknown (not yet tested) (ngs_pending)
- **Efficacy/toxicity weight:** 0.5

8 ranked therapeutic options across 5 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Idh1 R132 interventions

1. Ivosidenib (if IDH1 R132 confirmed on tumor NGS) *(conditional on idh1_r132 positive)*

Likelihood of effect: Moderate for control, low for shrinkage: randomized PFS HR 0.37 (pmid:32416072) but ORR only 2% and ITT OS non-significant (HR 0.69, p=0.060).

Toxicity burden: Low (ascites 7%, QT prolongation, fatigue — grade 3+ tracked placebo)

Counter-productive MoA:Low (No counter-productive vector; risktaker's dissent was magnitude-of-benefit, not mechanism.)

Overall:The only randomized and only category-1 option, but the survival claim rests on a PFS win and crossover modeling — buys control, not response.

Key references: PMID 32416072 · PMID 34848557 · NCT02989857

Evidence in detail

Ivosidenib (if IDH1 R132 confirmed on tumor NGS)

****Abou-Alfa et al. *Lancet Oncology* 2020****

- **Disease context:**Previously treated advanced cholangiocarcinoma with IDH1 R132 mutation (2L+) — IDH1 R132-mutant CCA after 1-2 prior regimens; crossover from placebo to ivosidenib permitted on progression
- **n:** 185
- **Toxicities:**Ascites (grade ≥ 3): 7.0% (n=121); **Fatigue** (grade ≥ 3): 3.0% (n=121); **Nausea** (grade ≥ 3): 2.0% (n=121); **QT prolongation** (grade 3): 1.0% (n=121); **Serious AE (any)** (grade any): 30.0% (n=121)
- **Efficacy:**HR_PFS 0.37 HR (95% CI 0.25-0.54); **PFS** 2.7 months (95% CI 1.6-4.2); **PFS_rate** 32.0% (95% CI 23.0-42.0); **ORR** 2.0%; **DCR** 53.0%; **HR_OS** 0.69 HR (95% CI 0.44-1.1); **HR_OS** 0.46 HR (95% CI 0.28-0.75)

Fgfr2 Fusion interventions

1. Futibatinib (if FGFR2 fusion/rearrangement confirmed) *(conditional on fgfr2_fusion positive)*

Likelihood of effect: Moderate-to-high in FGFR2-fusion disease: ORR 42%, mOS 21.7 mo (pmid:36652354), concordant in direction with FIGHT-202 and the selective agents.

Toxicity burden: Moderate (hyperphosphatemia 30% G3+, AST rise, stomatitis, fatigue)

Counter-productive MoA:Low (On-target FGFR inhibition; acquired kinase-domain resistance blunts durability but covalent binding delays escape.)

Overall:The published, on-label FGFR2 anchor with a resistance-coverage edge for sequencing; response is a surrogate endpoint with no randomized comparator.

Key references: PMID 36652354 · PMID 31109923 · NCT02052778

2. Pemigatinib (if FGFR2 fusion/rearrangement confirmed) *(conditional on fgfr2_fusion positive)*

Likelihood of effect: Moderate in FGFR2-fusion disease: ORR 35.5% (pmid:32203698), concordant with futibatinib in direction; single-arm surrogate endpoint.

Toxicity burden: High (grade 3+ AE 64%; hyperphosphatemia 55%, retinal detachment, hypophosphatemia)

Counter-productive MoA:Low (On-target FGFR inhibition; FGFR1-driven hyperphosphatemia is on-mechanism; acquired resistance mutations blunt durability.)

Overall:The longest-tenured FGFR2 option with the deepest safety record; edged below futibatinib on resistance-coverage sequencing, not on efficacy.

Key references: PMID 32203698 · NCT02924376

3. Lirafugratinib (RLY-4008), FGFR2-selective, via the ReFocus cohort (if FGFR2 fusion/rearrangement confirmed) *(conditional on fgfr2_fusion positive)*

Likelihood of effect: Moderate-to-high in FGFR2-fusion disease but low-certainty: ORR 46.5% (doi:10.1200/JCO.2026.44.2_suppl.476) is single-arm, abstract-only, pre-approval.

Toxicity burden: Moderate (palmar-plantar erythrodysesthesia 32.8% G3+, stomatitis 12.1%)

Counter-productive MoA: **Moderate** (FGFR2-selective on-target inhibition; critic and conservative dissented on evidence maturity and the 32.8% hand-foot burden, not a counter-productive mechanism per se.)

Overall: The highest-ORR FGFR2 option and the best hyperphosphatemia profile, but abstract-only, pre-approval, and carrying a one-in-three grade-3 hand-foot rate.

Key references: doi:10.1200/JCO.2026.44.2_suppl.476 · NCT04526106

Evidence in detail

Futibatinib (if FGFR2 fusion/rearrangement confirmed)

Goyal et al. *New England Journal of Medicine* 2023

- **Disease context:** Previously treated advanced intrahepatic cholangiocarcinoma with FGFR2 fusion or rearrangement (2L+) — Centrally confirmed FGFR2 fusion/rearrangement in intrahepatic CCA, ≥1 prior line; activity retained against some pemigatinib-resistance kinase-domain mutations
- **n:** 103
- **Toxicities:** Hyperphosphatemia (grade ≥3): 30.0% (n=103); Increased AST (grade ≥3): 7.0% (n=103); Stomatitis (grade ≥3): 6.0% (n=103); Fatigue (grade ≥3): 6.0% (n=103); Treatment-related discontinuation (grade any): 2.0% (n=103)
- **Efficacy:** ORR 42.0% (95% CI 32.0-52.0); DoR 9.7 months; PFS 9.0 months; OS 21.7 months

Pemigatinib (if FGFR2 fusion/rearrangement confirmed)

Abou-Alfa et al. *Lancet Oncology* 2020

- **Disease context:** Previously treated advanced/metastatic cholangiocarcinoma with FGFR2 fusion or rearrangement (2L+) — FGFR2 fusion/rearrangement cohort (cohort A), prior gemcitabine-based chemotherapy; ECOG 0-2
- **n:** 107
- **Toxicities:** Treatment-related grade 3+ AE (any) (grade ≥3): 64.0% (n=146); Hyperphosphatemia (grade any): 55.0% (n=146); Hypophosphatemia (grade ≥3): 12.0% (n=146); Alopecia (grade any): 46.0% (n=146); Dysgeusia (grade any): 38.0% (n=146); Nail toxicity (grade any): 42.0% (n=146); Arthralgia (grade ≥3): 6.0% (n=146); Stomatitis (grade ≥3): 5.0% (n=146); Palmar-plantar erythrodysesthesia (grade any): 11.0% (n=146); Serous retinal detachment (grade any): 4.0% (n=146)
- **Efficacy:** ORR 35.5% (95% CI 26.5-45.4); DCR 82.0% (95% CI 74.0-89.0); DoR 7.5 months (95% CI 5.7-14.5); PFS 6.9 months (95% CI 6.2-9.6); OS 21.1 months

ErbB2 Amp interventions

1. Zanidatamab (if HER2 IHC 3+ with reflex ISH confirmed) *(conditional on erbb2_amp positive)*

Likelihood of effect: Moderate in HER2-positive disease: confirmed ORR 41.3%, mDoR 12.9 mo (pmid:37276871), single-arm pivotal cohort with no randomized comparator.

Toxicity burden: Low (diarrhea 5% G3, decreased ejection fraction 3% G3, infusion reactions)

Counter-productive MoA:Low (On-target HER2 blockade; no mechanism-level dissent — board tension was T-DXd's ILD, resolved by ranking zanidatamab ahead.)

Overall:The dedicated-trial HER2 lead, ranked above T-DXd on a 10.5% ILD signal with no pulmonary baseline on file; gated on IHC 3+ with reflex ISH.

Key references: PMID 37276871 · PMID 36914633 · NCT04466891

2. Trastuzumab deruxtecan (T-DXd) (if HER2 IHC 3+/expressing confirmed) *(conditional on erbb2_amp positive)*

Likelihood of effect: High in IHC 3+ subgroup (ORR 56-61%) but low-certainty overall: full biliary-cohort ORR 22%, investigator-assessed (pmid:37870536), gated by ILD risk.

Toxicity burden: High (ILD/pneumonitis 10.5% with 3 treatment-related deaths; grade 3+ AE 40.8%)

Counter-productive MoA:Moderate (On-mechanism ILD/pneumonitis from the deruxtecan payload can cause fatal pulmonary injury, blunting the therapeutic goal in an unmonitored patient.)

Overall:Competitive IHC 3+ activity held below zanidatamab on a 10.5% ILD signal with three deaths and no pulmonary baseline; re-entry needs an ILD-monitoring plan.

Key references: PMID 37870536 · NCT04482309

Evidence in detail

Zanidatamab (if HER2 IHC 3+ with reflex ISH confirmed)

Harding et al. *Lancet Oncology* 2023

- **Disease context:**Previously treated HER2-amplified biliary tract cancer (2L+) — Cohort 1: HER2 amplified by ISH plus IHC 2+/3+ (4B5 companion antibody); prior gemcitabine-based therapy
- **n:** 80
- **Toxicities:**Diarrhea (grade any): 40.0% (n=80); Diarrhea (grade 3): 5.0% (n=80); Infusion-related reaction (grade any): 35.0% (n=80); Decreased ejection fraction (grade 3): 3.0% (n=80)
- **Efficacy:**ORR 41.3% (95% CI 30.4-52.8); DoR 12.9 months; PFS 5.5 months (95% CI 3.7-7.2)

Trastuzumab deruxtecan (T-DXd) (if HER2 IHC 3+/expressing confirmed)

Meric-Bernstam et al. *Journal of Clinical Oncology* 2024

- **Disease context:**HER2-expressing biliary tract cancer (pan-tumor basket cohort) (2L+) — Biliary tract cohort of a HER2-expressing solid-tumor basket; central HER2 IHC 3+ or 2+. IHC 3+ subgroup responded markedly better than the overall cohort
- **n:** 41

- **Toxicities:**Drug-related grade 3+ AE (any) (grade ≥ 3): 40.8%; ILD/pneumonitis (grade any): 10.5%; ILD/pneumonitis (grade 5); Nausea (grade any)
- **Efficacy:**ORR 22.0% (95% CI 10.6-37.6); **ORR** 61.3%; **ORR** 37.1%; **PFS** 6.9 months; **OS** 13.4 months

Braf V600E interventions

1. Dabrafenib + trametinib (if BRAF V600E confirmed on NGS/PCR) *(conditional on braf_v600e positive)*

Likelihood of effect: Moderate-to-high in BRAF V600E disease: ORR 47%, mPFS 9.0 mo (pmid:32818466); small single-arm cohort, low prevalence caps expected yield.

Toxicity burden: Moderate (pyrexia 19%, GGT rise 12% G3+, serious AE 40%)

Counter-productive MoA:Low (On-target MAPK inhibition; MEK co-inhibition mitigates paradoxical RAF activation. No mechanism-level dissent raised.)

Overall:A high-response, dual-endorsed doublet if V600E confirms; ranked low on 1-5% prevalence and a small single-arm cohort, not on any objection.

Key references: PMID 32818466 · NCT02034110

Evidence in detail

Dabrafenib + trametinib (if BRAF V600E confirmed on NGS/PCR)

Subbiah et al. *Lancet Oncology* 2020

- **Disease context:**BRAF V600E-mutated biliary tract cancer (2L+) — Molecularly confirmed BRAF V600E (NGS/PCR; VE1 IHC suggestive but not sufficient); non-V600E class 2/3 BRAF alterations do not qualify
- **n:** 43
- **Toxicities:**Serious AE (any) (grade any): 40.0% (n=43); **Treatment-related serious AE** (grade any): 21.0% (n=43); **Pyrexia** (grade any): 19.0% (n=43); **Increased gamma-glutamyltransferase** (grade ≥ 3): 12.0% (n=43)
- **Efficacy:**ORR 47.0% (95% CI 31.0-62.0); **DoR** 8.9 months; **PFS** 9.0 months (95% CI 5.0-10.0); **OS** 13.5 months (95% CI 10.0-33.0)

Msi Tmb interventions

1. Pembrolizumab (if MSI-H/dMMR or TMB-high confirmed) *(conditional on msi_tmb positive)*

Likelihood of effect: High and durable when MSI-H/dMMR positive (ORR 34.3%, mOS 23.5 mo, pmid:31682550) but low prevalence (1-3%); TMB-high a weaker predictor.

Toxicity burden: Low (immune-mediated AEs; grade 3-5 14.6%, one treatment-related death)

Counter-productive MoA:Low (T-cell exhaustion is a theoretical checkpoint risk but not raised by any persona for this biomarker-selected setting.)

Overall:Durable benefit when MSI-H/dMMR confirms, resolved free by the rank-1 workup; ranked last on 1-3% prevalence and no published CCA subgroup ORR.

Key references: PMID 31682550 · PMID 32919526 · NCT02628067

Evidence in detail

Pembrolizumab (if MSI-H/dMMR or TMB-high confirmed)

****Marabelle et al. *Journal of Clinical Oncology* 2020****

- **Disease context:MSI-H/dMMR advanced noncolorectal solid tumors (incl. biliary) (2L+)** — MSI-H/dMMR by PCR or IHC; 27 tumor types, cholangiocarcinoma among the most common. No CCA-specific subgroup ORR published
- **n:** 233
- **Toxicities:Treatment-related grade 3-5 AE (any)** (grade ≥ 3): 14.6% (n=233); **Treatment-related AE (any)** (grade any): 64.8% (n=233); **Pneumonia** (grade 5): (n=233)
- **Efficacy:ORR** 34.3% (95% CI 28.3-40.8); **PFS** 4.1 months (95% CI 2.4-4.9); **OS** 23.5 months; **TRAE_rate** 64.8%

****Marabelle et al. *Lancet Oncology* 2020****

- **Disease context:TMB-high advanced solid tumors (≥ 10 mut/Mb), incl. biliary (2L+)** — tTMB-high ≥ 10 mut/Mb on FoundationOne CDx; biliary admitted through the biomarker, not a dedicated cohort
- **n:** 102
- **Toxicities:** Safety mirrors the broader KEYNOTE-158 pembrolizumab population; this biomarker analysis did not report separate grade 3+ rates for the TMB-high subgroup.
- **Efficacy:ORR** 29.0% (95% CI 21.0-39.0); **DoR** not reached months; **PFS** 2.1 months (95% CI 2.1-4.1); **OS** 11.7 months (95% CI 9.1-19.1)